
Shipping Containers and Global Supply Chains

Stephen Weymouth
Dean's Professor

GEORGETOWN
UNIVERSITY

Module 3: what makes international (“cross-border”) business costly?



Geographic friction

Today: containerization

Freight · time · handling · coordination



Border friction

Next: currencies · jurisdictions · trade rules

Exchange rates · legal systems · trade policy · political risk

Containerization cut geographic friction through common standards and complementary investment.

Kees Torn / Wikimedia Commons · U.S. Customs and Border Protection

Your IB Voice

YOUR IB VOICE

QUESTION

Levinson says the container's value lies in how it is used. What exactly did the box change, and why did its largest effects take a decade to appear? Defend your answer with evidence from the chapter.

10 minutes · Individual · Notes permitted

Commit to one causal explanation before we reconstruct the system.

What changed immediately when cargo entered a box?

What changed only after the surrounding system was rebuilt?

Before the Box

Cargo crossed the port one piece at a time



Each handoff consumed labor and time, exposed cargo to loss, and required another organization to coordinate.

The port was the expensive break in the journey.

For one shipment, ports consumed almost half the cost



A dozen vendors moved one truckload. The two ports alone cost \$1,163 of the \$2,386 total.

Levinson, *The Box*, Chapter 1, Table 1

Physical trade costs could exceed tariffs

12%

of US export value went to ocean
freight in 1961

7%

average US import tariff in the same year

■ High transport costs protected nearby producers and discouraged international sales.

The System, Not the Box

Fifty-eight boxes sailed from Newark in 1956



McLean separated the container from the truck chassis and designed a continuous handoff among truck, ship, and rail.

Ideal-X, 1956 · Maersk/SeaLand

One fitting made different networks compatible



A standard corner casting lets cranes, chassis, rail cars, and ships handle the same box.

Compatibility required agreement over dimensions, fittings, strength, identification, and operating rules.

ArnoldReinhold / Wikimedia Commons · CC BY-SA 4.0

A container port operates like a factory



CONTAINERSHIP · 1,400 FT



4 FOOTBALL FIELDS · 1,200 FT



9,000
forty-foot containers

20
crew

200 ft
crane height

24 hrs
in port

Panama · Course photograph

The box reduced five costs at once



Lower total landed cost

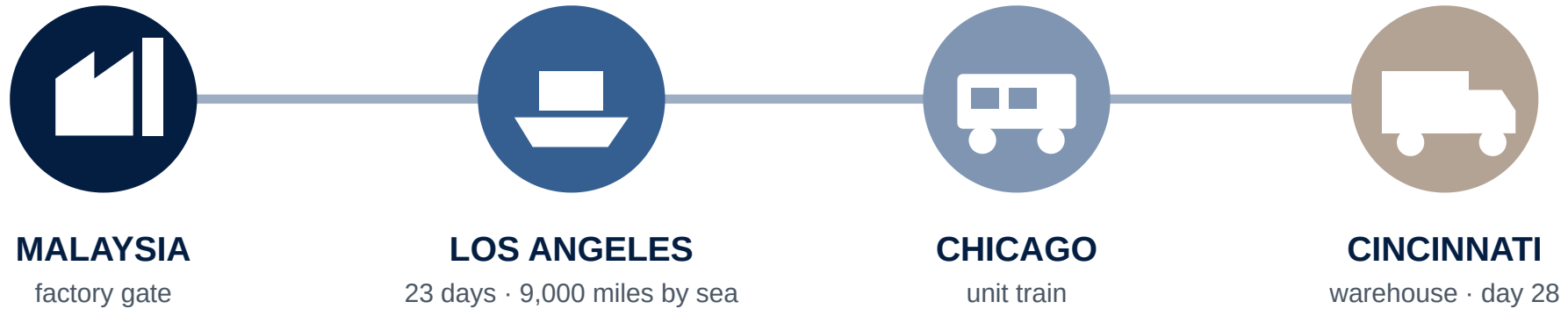
freight + insurance + inventory + delay + management

Reliability precise enough to schedule a delivery hour is what made just-in-time manufacturing practical at Toyota and Honda.

Lower freight bills explain only part of the economic change.

Malaysia to Cincinnati in 28 days, unopened

11,000 miles · 28 days · one sealed container



Glaeser and Kohlhase: it is now better to assume that moving goods is essentially costless than to assume it is an important component of production.

Why the payoff took a decade



Edison had the incandescent bulb by 1879. Twenty years later, 3 percent of American homes had electric light.

The technology worked in 1956. Its payoff waited for standards, infrastructure, organizational redesign, and political bargains.

Where Things Are Made

Ricardo assumed shipping was free

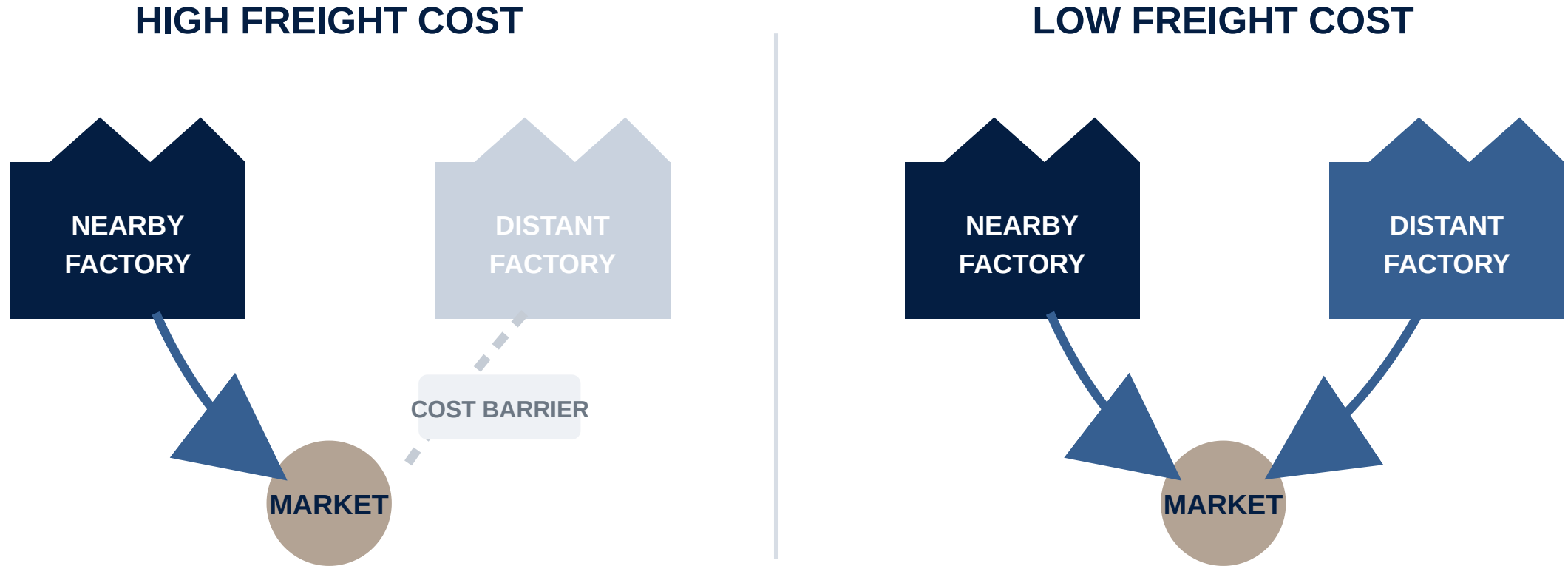
When Ricardo showed in 1817 that Portugal and England both gain by specializing, he counted only production costs. The cost of moving the wine and the cloth did not enter his analysis.

That assumption was carried into economists' models ever since. Serious study of how transport costs shape who makes what where began only in the 1990s.

As freight falls, plants move in sequence: first within a country, toward cheap land and low wages far from the ports, and only then abroad.

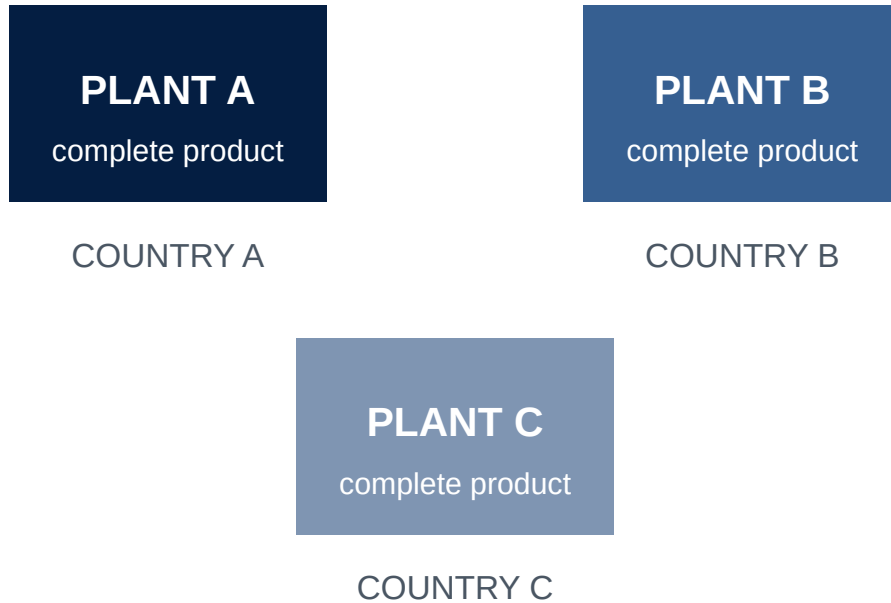
Comparative advantage was derived in a world without freight bills. For most of history, freight was the largest cost of all.

The global market came to domestic firms

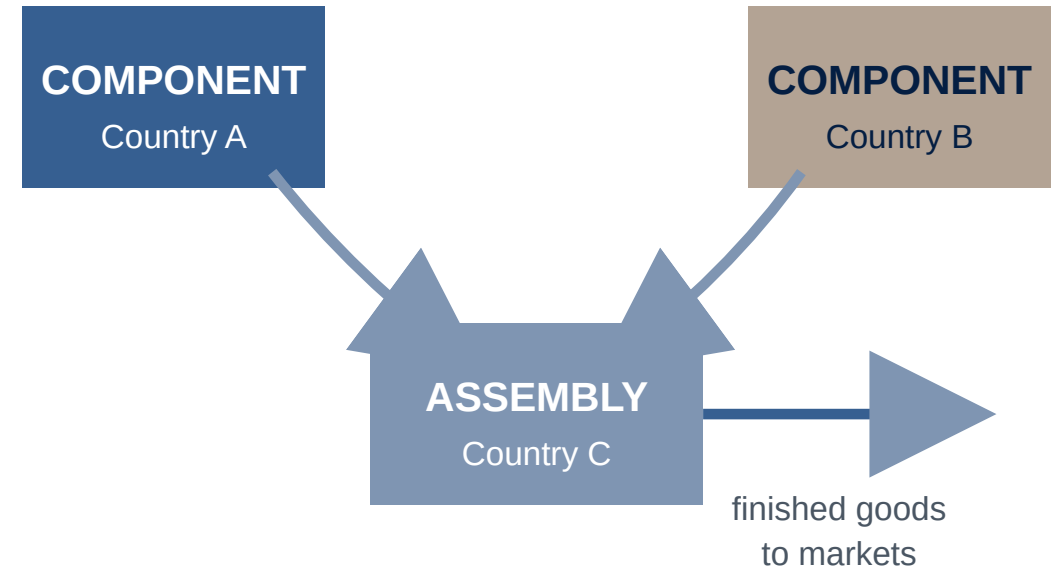


Multinational plants became one production network

SEPARATE NATIONAL PLANTS



ONE INTERNATIONAL NETWORK



Firms could reorganize production across viable locations as wages, exchange rates, capabilities, and other conditions changed.

Trade surged, but causation remains difficult

2x

manufactured trade grew more
than twice as fast as
manufacturing output after 1966

What else changed?

Tariffs · exchange rates · oil shocks · computers

How could we isolate the container's effect?

Propose a comparison that could separate the container from the other changes.

Workers occupied both sides of the ledger

As consumers

The United States imported four times as many product varieties in 2002 as in 1972.

Estimated benefit from added variety: nearly 3 percent of GDP.

As producers

New export centers gained factories, jobs, and investment.

Employers gained credible options to source or relocate abroad, exposing some workers to distant labor markets.

France ordered a shorter workweek with no cut in pay. Manufacturers avoided the cost by moving abroad.

■ The gains were broad. The losses concentrated in particular occupations, firms, and places.

Containerization relocated the port economy



Port Newark, 1996 · SCI Archives

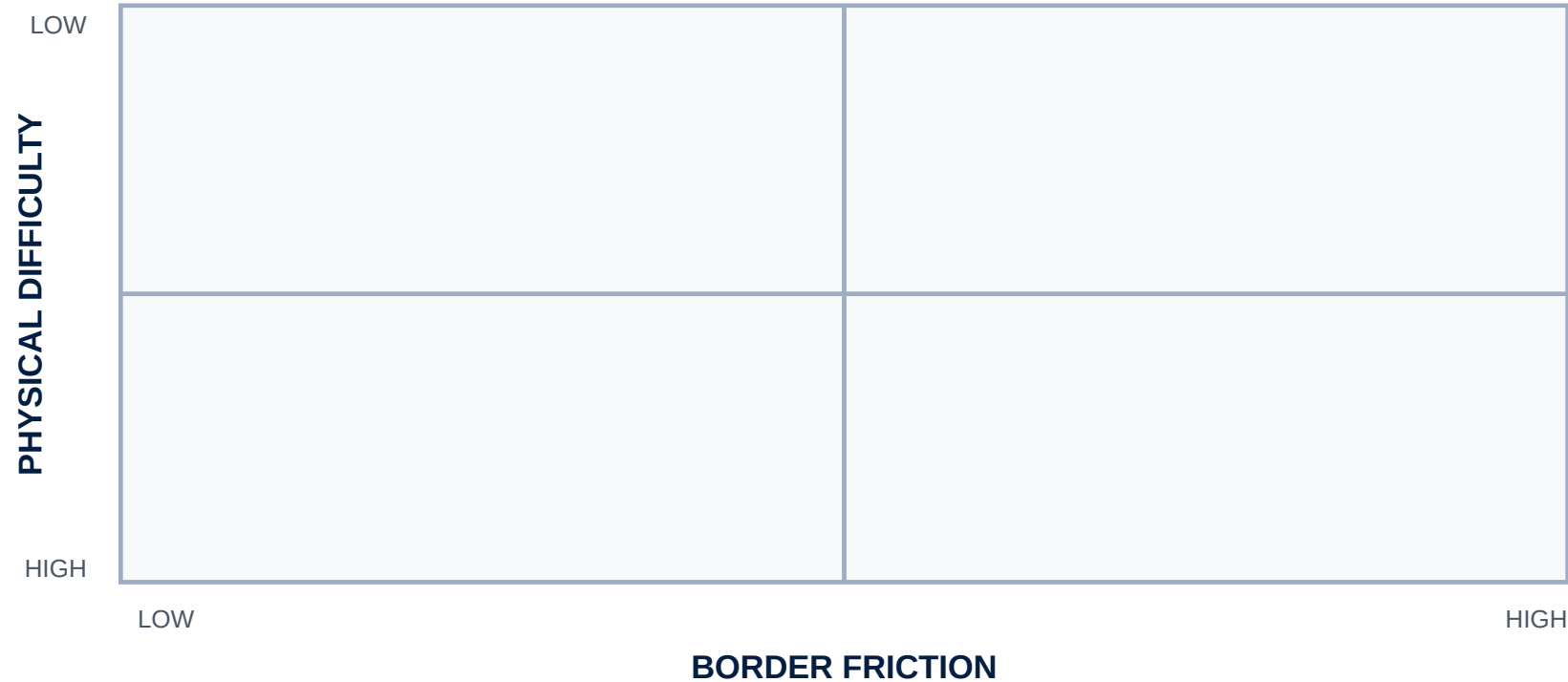
New York's finger piers lacked the contiguous land and surface connections container terminals required.

Port activity shifted toward Newark and Elizabeth.

Waterfront employment and investment moved with it.

Distance and Foreignness

Where do these products sit?



Running shoes · fresh flowers · a car · insulin · accounting software · a haircut · a smartphone · cement

Volunteers only. Place one product and name the cost that decides where it sits.

The sealed box is efficient for anyone

40,000

loaded containers crossing a large port
on an average working day

Every container carries a manifest.
Neither the ship line nor the port can
confirm that the manifest matches the
contents.

Opening the doors usually reveals a wall
of cardboard cartons.

| A system efficient for exporters is equally efficient for undeclared cargo.

Cheap freight did not erase location

Containerization changed

Handling · inventory · reliability ·
coordination

Borders still divide markets

Currencies · jurisdictions · trade rules ·
political risk

■ Why do firms keep production in high-wage locations?

Remove the Container

It is 1985. The container was never adopted.

The corner fitting was never agreed, the dock disputes were never settled, and cargo still crosses the pier one piece at a time.

Everything else is unchanged: jet travel, computers, the oil shocks, decolonization, Japanese manufacturing, and the same tariff reductions.

■ Does your case still happen in this world?

Your case

1. Japanese cars sold in the United States
2. Just-in-time manufacturing at Toyota
3. Walmart's business model
4. Shenzhen as a manufacturing center
5. The decline of the New York garment district
6. The Nespresso capsule from our first session
7. Bangalore's software services industry
8. Cut flowers flown from Kenya to Amsterdam

10 minutes · Groups of four · One verdict, one mechanism, one sentence of evidence from Levinson.

Which of these did the container cause, which did it accelerate, and which would have happened regardless?

Each group reports its verdict and its mechanism. We sort them into three columns.

What exactly did the box change, and why did its largest effects take a decade to appear?

Revise your opening answer. Which link in your causal argument changed?

Standards, complementary investment, organizational change, and political bargains lowered total landed cost. Global competition and fragmented production followed.

Tomorrow: one production network, many currencies.